# Daniel López-Castaño

Ph.D. student, Johns Hopkins University, Baltimore MD, USA
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Applied mathematician specializing in statistical models for structured high-dimensional data and machine learning. My work draws on geometry and probability, including optimal transport and manifold methods. I love translating theory into practical implementation.

## EDUCATION

**Johns Hopkins University**, Baltimore MD, USA — August 2021 — Present
Ph.D. in Applied Mathematics and Statistics
*Supervisors*: Mateo Díaz, Soledad Villar

**Johns Hopkins University**, Baltimore MD, USA — August 2021 — May 2023
M.Eng. in Applied Mathematics and Statistics
*Relevant Coursework*: Machine Learning 1, Machine Learning 2, Optimal Transport, Equivariant Machine Learning, Riemannian Geometry, Software Engineering for Data Science, Topics in Computer Graphics

**National University of Colombia**, Bogotá, Colombia — January 2018 — May 2020
M.Sc. in Mathematics
*Supervisors*: Sylvie Paycha (University of Postdam), Carolina Neira Jiménez

**National University of Colombia**, Bogotá, Colombia — January 2013 — December 2017
B.Sc. in Mathematics
*Supervisor*: Carolina Neira Jiménez

## RELEVANT EXPERIENCE

**Inter-American Development Bank - Risk Management** — Washington DC, USA
*Systemic Risk Research Intern* — Jun 2026 — August 2026
Developed a systemic risk forecasting framework using graph autoregression to capture temporal dependencies and network propagation, translating interconnected risk signals into institutional exposure estimates. Delivered a risk monitoring system.

**The Johns Hopkins University - Applied Physics Laboratory** — Baltimore MD, USA
*WSE-APL Research Assistant* — May 2023 — August 2023
Developed a density estimation method grounded in universal approximation theory, using compact sigmoid–polynomial models to capture heavy-tailed and multimodal distributions with inference costs independent of dataset size.

**Vozy** — Bogotá, Colombia
*Machine Learning Engineer* — September 2020 — July 2021
Developed a voice biometrics system using PyTorch and TensorFlow that surpassed contemporary state-of-the-art benchmarks; deployed conversational AI through APIs and Docker, and mentored Holberton School interns in NLP research.

**University of Potsdam** — Potsdam, Germany
*Research Intern* — August 2019 — November 2019
Conducted research for master's thesis on the Cesàro theory of distributions and its asymptotic analysis, exploring asymptotic expansions of the spectral action in noncommutative geometry.

## TEACHING EXPERIENCE

**National University of Colombia** — Bogotá, Colombia
*Instructor* — August 2026

Delivered a minicourse on Geometric Data Science, covering optimization on manifolds, optimal transport, Wasserstein gradient flows, and applications.

**The Johns Hopkins University** Baltimore MD, USA
*Graduate Teaching Assistant* August 2023 — May 2025, and August 2026 — Present

Developed instructional materials for Machine Learning I and II and served as a teaching assistant for both courses and Nonlinear Optimization II, delivering supplementary lectures and review sessions.

**Instituto Universitario de Las Américas y El Caribe** Colima, Mexico
*Visiting Lecturer (Remote)* November 2019 — April 2020

Developed the syllabus and served as an instructor for (i) Arithmetic, Trigonometry, and its Concept Development, and (ii) Linear Algebra courses in the master's program in Mathematical Education.

**Universidad Nacional de Colombia** Bogotá, Colombia
*Lecturer and Graduate Assistant* August 2018 — December 2019

Instructor for Calculus 1 (Cálculo Diferencial 1000004-B) covering four groups of 40-60 students over 18 months.

**Fundación Círculo de Excelencia Académica** Bogotá, Colombia
*Mathematics Teacher* March 2013 — July 2018

Taught and advised students from vulnerable communities across Colombia, designing curricula and instructional materials for academic placement programs.

## PUBLICATIONS

**Working papers & preprints**

*Estimating Morse Information from Samples* (with M. Díaz, A. Quiroz, M. Velasco –in preparation)
Developed a nonparametric method with finite-sample guarantees for recovering the critical points and Morse indices of height functions on an unknown manifold from point-cloud data.

*Wasserstein approximation of measures via orthogonal frames: statistical and computational complexities* (with M. Díaz, N. García Trillos, C. Yang – alphabetical order), 2026 - arXiv:2610.36581
Established the parametric rate $n^{-1/2}$ for nonsmooth orthogonal-dictionary learning estimation for the empirical risk minimizer and Riemannian SGD.

*Inference-fast density estimation via universal approximators* (with J. Schmidt –in preparation)
Established mathematical foundations through universal approximation theorems, transforming density estimation into supervised learning framework. Research conducted at the Johns Hopkins Applied Physics Laboratory.

**Master's Thesis**

*A distributional approach to asymptotics of the spectral action* (2020) - repositorio.unal.edu.co/handle/unal/79294
Developed rigorous asymptotic analysis using Cesàro theory of distributions. Research conducted at Universität Potsdam under supervision of Prof. Sylvie Paycha.

**Undergraduate Thesis**

*Complex powers of elliptic pseudodifferential operators* (2017)
Reconstructed the proofs in Complex Powers of Pseudodifferential Operators, connecting the analytical framework to applications in index theory.

## SKILLS

**Programming:** Python (NumPy, Pandas, Scikit-learn, PyTorch, JAX), version control (Git), containerization (Docker), backend (Flask), frontend (HTML, Javascript), bash shell scripting

**Data visualization:** MANIM, Plotly, Seaborn, Matplotlib

**Software:** Postman, VSCode, GeoGebra, Excel

**Languages:** English (proficient), Spanish (native)

## PROJECTS

**Systemic Risk Framework** Washington DC, USA
*Researcher* Jun 2026 — Aug 2026

Developed a Python package for systemic risk forecasting on signed networks using graph autoregression, with reproducible pipelines, model benchmarks, automated tests, and diagnostics of institutional exposure and predictive performance.

**Vision Guided Drone** Baltimore MD, USA
*Mentor* Feb 2025 — May 2025

Mentored three high school interns at Johns Hopkins University in developing autonomous drone navigation, implementing computer vision and path optimization algorithms for efficient flight.

**Qudost** Baltimore MD, USA
*Research Assistant* May 2023 — May 2024

Developed a supervised density estimation method for real-time inference during a summer research internship at Johns Hopkins University Applied Physics Laboratory.

**VoBio** Bogotá, Colombia
*Machine Learning Engineer* September 2020 — July 2021

Integrated voice biometrics into client's authentication systems, improving user experience and operational workflows.

**LILI-STT** Bogotá, Colombia
*Machine Learning Engineer* September 2020 — July 2021

Trained domain-specific automatic speech recognition models on GPUs with the NVIDIA NeMo framework.

## HONORS

**Fulbright Scholarship Minciencias** Baltimore MD, USA
Doctoral scholarship for studies in the U.S. August 2020

**Instructor scholarship *Beca Asistente Docente*** Bogota, Colombia
Graduate instructor assistanship at National University of Colombia. August 2018 - February 2020

**Honors Tuition** Bogotá, Colombia
Scholarship for the highest entrance exam score (2013-1) and top grades (2013-2). February 2013 - December 2013

## FELLOWSHIPS

**Acheson J. Duncan Research Fund** Baltimore MD, USA
Travel grant to present at the INABAG School+Conference in Siena, Italy. August 2026

**Acheson J. Duncan Research Fund** Baltimore MD, USA
Travel grant to attend the Summer School on Optimal Transport, Stochastic Analysis, and Applications to Machine Learning in Daejeon, South Korea. June 2024

**UNM-NSF** Guanajuato, Mexico
Travel grant to attend the 2023 Workshop on Potential Theory at CIMAT-Guanajuato. September 2023

**INMAS** Baltimore MD, USA
Internship training and financial aid for students in the field of mathematical sciences. May 2023

**IAMP** Potsdam, Germany
Travel grant to attend a summer school in Noncommutative Manifolds and their symmetries in Scalea, Italy. September 2019

**UNI-GOTTINGEN** Göttingen, Germany
Funding to present a contributed talk at the Summer School on L2-Torsion and Symmetric Spaces. October 2019

**CIMPA** Mérida, Mexico
Funding to present at the CIMPA Research School on Noncommutative Geometry and Index Theory hosted at CIMAT-Mérida. November 2018

**UNIANDES** Villa de Leyva, Colombia
Funding to present a contributed talk at the Villa de Leyva Summer School 2017 on Geometric, Algebraic and Topological Methods for Quantum Field Theory. July 2017

## VOLUNTEER ACTIVITIES

**Whiting Internships in Science and Engineering (WISE) - Johns Hopkins** Baltimore MD, USA
*Mentor* February 2025 — May 2025

Mentored high school interns, encouraging STEM interest and skill development through hands-on guidance and collaborative problem-solving.

**Directed Reading Program - Johns Hopkins Chapter** Baltimore MD, USA
*Mentor* September 2023 — December 2024

*Fall 2024* - Geometry of the Space of Probability Distributions (Mentee: Shayaan Emran).
*Spring 2024* - Symmetries and Representations (Mentee: Minjae Kim).
*Fall 2023* - Spectral Geometry on Triangles (Mentee: Sukriti Gupta).

**FUNAMJE** Cota, Colombia
*Cultural Promoter* 2008 — July 2021

Volunteer work with the elderly. Collection and delivery of essential items and food.

## ACADEMIC TALKS

**INABAG School+Conference - University of Siena** Bogotá, Colombia
Short communication: *Estimating Morse indices from samples* September 2026

**UN Encuentro de Matematicas - National University of Colombia** Bogotá, Colombia
Plenary speaker: *Wasserstein approximation of measures via orthogonal frames* August 2026

**Statistics and Data Science Workshop - University of Los Andes** Bogotá, Colombia
Poster: *Estimating Morse information from samples* December 2025

**Optimization Workshop - University of Los Andes** Bogotá, Colombia
Poster: *Geometric Factor Analysis* - **Runner-up for Best Poster Award** December 2024

**Jr MINDS Seminar - Johns Hopkins University** Baltimore MD, USA
Poster: *Geometric Factor Analysis* November 2024

**SIAM Mathematics of Data Science** Atlanta GA, USA
Poster: *Geometric Factor Analysis* October 2024

**Johns Hopkins AI-X Foundry Fall 2023 Symposium** Baltimore MD, USA
Poster: *Inference-fast density estimation via universal approximators* September 2023

**Colloquium Mathematics Department - National University of Colombia** Bogotá, Colombia
Conference: *On vibrating strings, drums and what can and cannot be heard* December 2019

**Analysis Group Seminar - University of Potsdam** Potsdam, Germany
Conference: *On the distributional asymptotic expansion of the Spectral Action* November 2019

**Summer School on L2-Torsion and Symmetric Spaces - University of Göttingen** Göttingen, Germany
Short communication: *Sunada's technique for constructing isospectral manifolds* October 2019

**Analysis Group Seminar - University of Potsdam** Potsdam, Germany
Conference: *A space of symbols and the Cesàro behavior at infinity* September 2019

**Uniandes SRI-2019 on Geometry and Theoretical Physics** Villa de Leyva, Colombia
Short communication: *Notions of Spectral Action* July 2019

**CIMPA School on Noncommutative Geometry and Index Theory - CIMAT Mérida** Mérida, Mexico
Short communication: *Spectral Geometry and Spectral Action* December 2018

**Geometric, Algebraic and Topological Methods for Quantum Field Theory** Villa de Leyva, Colombia
Short communication: *Index formulas of an elliptic operator* July 2017

**TOP3-UD - Francisco José de Caldas District University** Bogotá, Colombia
Short communication: *The weak$^*$ topology on distributions and calculation rules* May 2017

**VII Encuentro Nacional de Matemáticas y Estadística - University of Tolima** Ibagué, Colombia
Short communication: *Radon measures: an approximation of topological dual of $L^\infty$* May 2017